GOA COLLEGE OF ENGINEERING

"BHAUSAHEB BANDODKAR TECHNICAL EDUCATION COMPLEX"
FARMAGUDI, PONDA- GOA - INDIA

MATHEMATICS-III
UNIT 1: MATRICES
LECTURE 1

DEVELOPED BY MATHEMATICS FACULTY
DEPARTMENT OF SCIENCE & HUMANITIES
GOA COLLEGE OF ENGINEERING

MATRICES LECTURE 1

TOPICS TO BE COVERED:

- INTRODUCTION TO MATRICES
- TYPES OF MATRICES
- MATRIX OPERATIONS
- PROPERTIES OF MATRIX OPERATIONS

Introduction To Matrices:

Definition: A matrix is a rectangular array of objects (real or complex numbers, variables, mathematical expressions or functions).

i.e typically a system of mn numbers which are arranged in m-rows and n-columns.

Matrices are denoted by capital letters like A, B, C etc.

If matrix has 'm'-rows and 'n'-columns then the order of the matrix is "m x n".

A general m x n matrix is represented as $A=\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$

In mathematical notation A=$[a_{ij}]_{m\,x\,n}$, *where* a_{ij} denotes the element in the i^{th} *row and* j^{th} column.

a_{11} , a_{22}, a_{33} , *etc are* called the diagonal elements of the matrix

Examples of Matrices:

1. $A = \begin{bmatrix} 1 & 2 & 5 \\ 0 & 3 & 2 \\ 1 & 5 & 3 \end{bmatrix}$ is a matrix of order 3 x 3.

In this matrix, $a_{11} = 1, a_{12} = 2, a_{13} = 5\ etc$. The rows are [1 2 5] the first row R_1, [0 3 2] the second row R_2 and $[1\ 5\ 3]$the third row R_3. The columns are $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ the first column C_1, $\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$ the second column C_2 and $\begin{bmatrix} 5 \\ 2 \\ 3 \end{bmatrix}$ the third column C_3.

2. $B = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ is a matrix of order 2 x 2

3. $C = \begin{bmatrix} 2 & 3 \\ 3 & 2 \\ 1 & 5 \end{bmatrix}$ is a matrix of order 3 x 2

4. $D = \begin{bmatrix} 5 & 3 & 2 \\ 6 & 0 & 1 \end{bmatrix}$ is a matrix of order 2 x 3

Some Applications of Matrices:

1. Solving system of Equations: Matrices are used to solve system of linear equations. Cramer's rule is used for this purpose.
2. Graph theory: A matrix is a convenient and useful way of representing a graph in a computer.
3. Computer Graphics & Image processing : Graphic software such as Adobe Photoshop on your personal computer uses matrices to process linear transformations to render images. A square matrix can represent a linear transformation of a geometric object. Matrices are used to project three dimensional images into two dimensional planes
4. Information Technology: Many IT companies also use matrices as data structures to track user information, perform search queries, and manage databases. Matrices are used in the compression of electronic information
5. Cryptography: Matrices are used to write, encode, decode and send secret messages.
6. Economics: Matrix Cramer's Rule and determinants are simple and important tools for solving many problems in business and economics related to maximize profit and minimize loss
7. Physics: Matrices are used in science of optics to account for reflection and for refraction. Matrices are also useful in electrical circuits and quantum mechanics and resistor conversion of electrical energy. Matrices are used to solve AC network equations in electric circuits.
8. Geology: Matrices are used for making seismic surveys.

Application of Matrices in Mechanical Engineering

What is the use of matrices in mechanical engineering?

- Strength of materials - Strain matrix, stress matrix and the moment of inertia tensor. These strain, stress matrices have to be manipulated by using theory of matrices.
- Engineering Material Sciences - Miller indices. Used for defining crystal lattice geometries.
- Computer Aided Designing (CAD) - Without matrices, this subject cannot exist.

Matrices are used for noting down all the joint variables for forward/inverse kinematics and dynamics problems of the subject. Finite Element Analysis (FEA) and Finite Element Methods (FEM) - This subject **uses** many concepts of **matrices** for solving problems, just like CAD does.

Types of Matrices:

1. **Row matrix** : A matrix which has one row and n-numbers of columns.

   The general form of a row matrix of order "1 x n" is $[a_{11} \quad a_{12} \quad a_{13} \cdots \quad a_{1n}]$.

   Ex. $A = [1 \quad 3 \quad 5 \quad 8]$ is a row matrix having order "1 x 4".

2. **Column matrix**: A matrix having one column and m-number of rows.

   The general form of a column matrix of order "m x 1" is $\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}$.

   Ex. B= $\begin{bmatrix} 5 \\ 6 \\ 2 \end{bmatrix}$ is a column matrix of order "3 x 1".

Types of matrix:

3. **Rectangular matrix**: A matrix $A = [a_{ij}]_{m \times n}$ is called a rectangular matrix if "m ≠ n" .

Ex $A = \begin{bmatrix} -1 & 2 & 5 & 4 \\ 3 & 0 & 6 & 8 \\ -5 & -4 & 1 & 0 \end{bmatrix}_{3x4}$

4. Square Matrix: A matrix $A = [a_{ij}]_{m \times n}$ is called a square matrix if "m = n" .
i.e A matrix which has same number of rows and columns.

Ex A= $\begin{bmatrix} 1 & 5 & 7 \\ 0 & 2 & 3 \\ 5 & 3 & 3 \end{bmatrix}_{3x3}$ $and \quad B = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix}_{2x2}$

a_{11}= 1, a_{22} =2 and $a_{33} = 3$ are called the principal or leading diagonal elements.

The sum of diagonal elements is called the trace of a matrix and denoted by trace(A).

In the above example trace(A)=1+2+3=6

Types of matrix:

5. Null or Zero matrix: A matrix A=$[a_{ij}]_{m \times n}$ is called a null matrix if all the elements are zero. It is denoted by O.

$$a_{ij} = 0 \,, \forall i, j$$

Examples: $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3x3}$ $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{2x3}$

6. Diagonal matrix: A square matrix where all the elements are zero except those on the leading diagonal.

A square matrix A=$[a_{ij}]_{n \times n}$ is called a diagonal matrix if a_{ij} =0 for all $i \neq j$

Examples: $\begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3x3}$ $\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -9 \end{bmatrix}_{4x4}$

Types of matrix:

7. Scalar matrix: A square matrix A=$[a_{ij}]_{n \times n}$ is called a scalar matrix if

$$a_{ij} = 0 \ , \ i \neq j$$
$$= c \ , \ i = j$$

i.e. a diagonal matrix whose diagonal elements are equal to a constant c.

Examples: $\begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ $\begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$

8. Identity or Unit matrix: : A square matrix A=$[a_{ij}]_{n \times n}$ is called an identity matrix if

$$a_{ij} = 0 \ , \ i \neq j$$
$$= 1, \ i = j$$

i.e. a scalar matrix whose leading diagonal elements are equal to 1. It is denoted by I_n.

Examples: $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Types of matrix:

9. Upper triangular matrix: A square matrix A=$[a_{ij}]_{n\,\mathrm{x}\,n}$ is called an upper triangular matrix if all the elements below the principal diagonal are zero.

Example: $\begin{bmatrix} -2 & 1 & -3 \\ 0 & 5 & 4 \\ 0 & 0 & 7 \end{bmatrix}$

$$a_{ij} = 0 \ , \ i > j$$

10. Lower triangular matrix: A square matrix A=$[a_{ij}]_{n\,\mathrm{x}\,n}$ is called a lower triangular matrix if all the elements above the principal diagonal are zero.

$$a_{ij} = 0 \ , \ i < j$$

Examples: $\begin{bmatrix} 3 & 0 & 0 \\ -5 & 1 & 0 \\ -1 & 0 & 8 \end{bmatrix}$

11. Triangular matrix: A square matrix A=$[a_{ij}]_{n\,\mathrm{x}\,n}$ is called a triangular matrix if it is upper as well as lower triangular matrix.

$$a_{ij} = 0 \ , \ i \neq j$$

i.e. it is a diagonal matrix

Equality of Matrices

Definition: Two matrices A and B are said to be equal if

i. they are of same order.

ii. all corresponding elements are equal.

Let $A = [a_{ij}]_{mxn}$ and $B = [b_{ij}]_{mxn}$

If $A = B$, then $a_{ij} = b_{ij} \; \forall \, i, j$

Some properties of equality:

- If **A** = **B**, then **B** = **A** for all **A** and **B**
- If **A** = **B**, and **B** = **C**, then **A** = **C** for all **A**, **B** and **C**

Operations on Matrices

1. Addition and Subtraction : Let $A=[\ a_{ij}\ ]_{m\times n}$ and $B=[\ b_{ij}\ ]_{m\times n}$. Then

- Sum $A + B =[\ a_{ij} + b_{ij}\ ]_{m\times n}$
- Difference $A - B = [\ a_{ij} - b_{ij}\ ]_{m\times n}$

Note: Matrices of different orders cannot be added or subtracted.

Examples:

1. $$\begin{bmatrix} 4 & -3 & -1 \\ 2 & 6 & 5 \end{bmatrix}_{2x3} + \begin{bmatrix} 1 & 2 & 5 \\ -2 & -2 & 3 \end{bmatrix}_{2x3} = \begin{bmatrix} 5 & -1 & 4 \\ 0 & 4 & 8 \end{bmatrix}_{2x3}$$

2. $$\begin{bmatrix} 4 & -3 & -1 \\ 2 & 6 & 5 \end{bmatrix}_{2x3} - \begin{bmatrix} 1 & 2 & 5 \\ -2 & -2 & 3 \end{bmatrix}_{2x3} = \begin{bmatrix} 3 & -5 & -6 \\ 4 & 8 & 2 \end{bmatrix}_{2x3}$$

Properties of Matrix Addition

Properties

- Commutative Law: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
- Associative Law: $\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + \mathbf{B} + \mathbf{C}$
- Existence of additive identity: $\mathbf{A} + \mathbf{0} = \mathbf{0} + \mathbf{A} = \mathbf{A}$
- Existence of additive inverse $\mathbf{A} + (-\mathbf{A}) = \mathbf{0}$ (where $-\mathbf{A}$ is the matrix composed of $-a_{ij}$ as elements)

Operations on Matrices

2. Scalar multiplication : Let $A=[\,a_{ij}\,]_{m \times n}$ and $k \neq 0$ be any scalar. Then

$$kA = [\,ka_{ij}]_{m \times n}$$

i.e. Each element in the matrix is multiplied by the scalar k.

Examples: $If\ A = \begin{bmatrix} 4 & -3 & -1 \\ 2 & 6 & 5 \end{bmatrix}_{2x3}, then$

1. $$2A = \begin{bmatrix} 8 & -6 & -2 \\ 4 & 12 & 10 \end{bmatrix}_{2x3}$$

2. $$-5A = \begin{bmatrix} -20 & 15 & 5 \\ -10 & -30 & -25 \end{bmatrix}_{2x3}$$

Properties of Scalar Multiplication

- $k\,(\mathbf{A} + \mathbf{B}) = k\mathbf{A} + k\mathbf{B}$
- $(k + l)\mathbf{A} = k\mathbf{A} + l\mathbf{A}$
- $k(\mathbf{AB}) = (k\mathbf{A})\mathbf{B} = \mathbf{A}(k\mathbf{B})$
- $k(l\mathbf{A}) = (kl)\mathbf{A}$

Operations on Matrices

3. Multiplication:

- Two matrices **A** and **B** are said to be **conformable** for multiplication if the number of columns in A equals the number of rows in B.

- $\mathbf{A}_{m \times n} \;\; \times \;\; \mathbf{B}_{n \times k} \;\; = \;\; \mathbf{C}_{m \times k}$

- If $A=[a_{ij}]_{m \times n}$ and $B=[b_{jk}]_{n \times k}$, the product matrix $C=AB=[c_{ik}]_{mxk}$ has order m x k and each element in C can by computed by:

$$c_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk}$$

Operations on Matrices

Multiplication rule: The ij^{th} element of the product matrix is obtained by multiplying each element in the i^{th} row of the first matrix by the corresponding element in the j^{th} column of the second matrix and adding the products.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}_{2\text{x}3} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}_{3\text{x}2} = \begin{bmatrix} a_{11}b_{11}+a_{12}b_{21}+a_{13}b_{31} & a_{11}b_{12}+a_{12}b_{22}+a_{13}b_{32} \\ a_{21}b_{11}+a_{22}b_{21}+a_{23}b_{31} & a_{21}b_{12}+a_{22}b_{22}+a_{23}b_{32} \end{bmatrix}_{2\text{x}2}$$

Example 1:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 2 & 7 \end{bmatrix}_{2\text{x}3} \cdot \begin{bmatrix} 4 & 8 \\ 6 & 2 \\ 5 & 3 \end{bmatrix}_{3\text{x}2} = \begin{bmatrix} 1\cdot 4+2\cdot 6+3\cdot 5 & 1\cdot 8+2\cdot 2+3\cdot 3 \\ 4\cdot 4+2\cdot 6+7\cdot 5 & 4\cdot 8+2\cdot 2+7\cdot 3 \end{bmatrix}_{2\text{x}2} = \begin{bmatrix} 31 & 21 \\ 63 & 57 \end{bmatrix}_{2\text{x}2}$$

$$\begin{bmatrix} 4 & 8 \\ 6 & 2 \\ 5 & 3 \end{bmatrix}_{3\text{x}2} \cdot \begin{bmatrix} 1 & 2 & 3 \\ 4 & 2 & 7 \end{bmatrix}_{2\text{x}3} = \begin{bmatrix} 4\cdot 1+8\cdot 4 & 4\cdot 2+8\cdot 2 & 4\cdot 3+8\cdot 7 \\ 6\cdot 1+2\cdot 4 & 6\cdot 2+2\cdot 2 & 6\cdot 3+2\cdot 7 \\ 5\cdot 1+3\cdot 4 & 5\cdot 2+3\cdot 2 & 5\cdot 3+3\cdot 7 \end{bmatrix}_{3\text{x}3} = \begin{bmatrix} 36 & 24 & 68 \\ 14 & 16 & 32 \\ 17 & 16 & 36 \end{bmatrix}_{3\text{x}3}$$

Matrix Multiplication

Example 2

- A=$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 0 \end{bmatrix}_{2\,x\,3}$ and B= $\begin{bmatrix} 1 & 5 & 1 \\ 2 & 2 & 3 \\ 2 & 0 & 1 \end{bmatrix}_{3\,x\,3}$

- Then C=AB= $\begin{bmatrix} 1.1+2.2+3.2 & 1.5+2.2+3.0 & 1.1+2.3+3.1 \\ 3.1+1.2+0.2 & 3.5+1.2+0.0 & 3.1+1.3+0.1 \end{bmatrix}_{2x\,3}$

$=\begin{bmatrix} 11 & 9 & 10 \\ 5 & 17 & 6 \end{bmatrix}$

BA does not exist.

Properties of Multiplication

Properties: Let matrices **A**, **B** and **C** be conformable for the operations indicated, then

1. A(BC)=(AB)C (Associative)
2. AI=A=IA (Existence of identity)
3. $A(B+C) = AB + AC$ (Distributive property)

Note:

- **AB ≠BA, In general matrix multiplication is non commutative**

$$Let\ A = \begin{bmatrix} 1 & 2 \\ 5 & 0 \end{bmatrix}\ and\ B = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 2 \\ 5 & 0 \end{bmatrix}\begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 15 & 20 \end{bmatrix}$$

$$BA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} 23 & 6 \\ 10 & 0 \end{bmatrix}$$

$$AB \neq BA$$

Properties of Multiplication

- $A \cdot B = 0 \implies A \neq 0$ and $B \neq 0$ (Existence of zero divisors)

Ex. $A = \begin{bmatrix} 3 & 2 \\ 3 & 2 \end{bmatrix}$ $B = \begin{bmatrix} -2 & 0 \\ 3 & 0 \end{bmatrix}$

$AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$ but $A \neq 0$ $B \neq 0$

- If **AB** = **AC**, **B** is not necessarily equal to **C**

- $(A + B)^2 = A^2 + AB + BA + B^2$

Practical Application

Suppose you are a business owner and sell clothing. The following represents the number of items sold and the cost for each item: We will use matrix operations to determine the total revenue over the two days:

Monday: 3 T-shirts at $10 each, 4 Trousers at $15 each, and 1 pair of shorts at $20.

Tuesday: 4 T-shirts at $10 each, 2 Trousers at $15 each, and 3 pairs of shorts at $20.

Represent the information using two matrices: The product of the two matrices give the total revenue:

Unit price of each item:

Total Revenue for Monday & Tuesday

$$\begin{bmatrix} 10 & 15 & 20 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 4 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 110 & 130 \end{bmatrix}$$

Qty sold of each item on Monday

Qty sold of each item on Monday

THANK YOU!